

and parliament, automatically raise aggregate expenditure in times of recession and reduce it in times of boom, thus ensuring economic stability. These fiscal measures are therefore called automatic stabilizers or built-in stabilizers. Since these stabilizers are built into the budget and do not require any deliberate action by the government, they represent non-discretionary fiscal policy. Built in stability of tax revenue and government expenditure of transfer payments and subsidies is created because they vary with national income. These taxes and transfers automatically bring about appropriate changes in the aggregate expenditure and reduce the impact of recession and inflation. This means that recession and inflation will be less intense with the operation of the automatic stabilizers. Let us understand how taxes and transfers work to reduce the fluctuations of business cycles.

18.4.1 Income Taxes

At given net tax rates, a fall in national income, output, and employment reduces tax collections. In other words, when national income declines at times of recession, the tax revenue declines as well. This prevents aggregate expenditure from falling by the same proportion as the decline in income because the disposable income changes by less than the change in national income. The slope of the aggregate expenditure function $(c(1-t)-m)$ is lower, and so is the multiplier. Conversely, in a boom, net tax revenues rise and disposable income rises by less than the rise in national income, which helps dampen the boom.

18.4.2 Transfer Payments

When there is recession and as a result unemployment increases, the Government has to spend more on compensation for unemployment and other welfare programmes such as food stamps, rent-subsidies, subsidies to farmers. This hike in Government expenditure tends to make recession short-lived and less intense. On the other hand, when at times of boom and inflation, national income increases and therefore unemployment falls, the Government curtails its programme of social benefits which results in lowering Government expenditure. This smaller government spending helps to rein in inflation.

18.4.3 Discretionary vs. Non-Discretionary Measures

It follows from the above discussion that automatic stabilizers reduce the intensity of business fluctuations that is, both recession and inflation. By reducing the sensitivity of the economy to expenditure shocks, automatic stabilizers are always at work reducing the size of output and employment fluctuations. However, automatic stabilizers only serve to moderate the fluctuations in real GDP caused by fluctuations in autonomous expenditure. They do not offset those autonomous expenditure disturbances. There is no automatic

change in autonomous government expenditure or tax rates. Those changes usually come from discretionary fiscal policy.

Fig 18.6 shows how the budget balance gets affected as a result of the operation of non-discretionary and discretionary fiscal policies. A government budget function $BB_0 = t_0Y + G_0$ and a structural budget balance SBB_0 at potential output (Y_f) is shown in Fig. 18.6. This budget function represents a fiscal program. Automatic stabilization is a part of this program. It comes from the *slope of the budget function*, the net tax rate t_0 in this case. Automatic stabilization works by changing the budget balance along the BB_0 line as income fluctuates between Y_1 and Y_2 . On the other hand, discretionary stabilization shifts the budget function as a result of changes in government expenditure or taxes.

Fig 18.6: Automatic and Discretionary Fiscal Policy

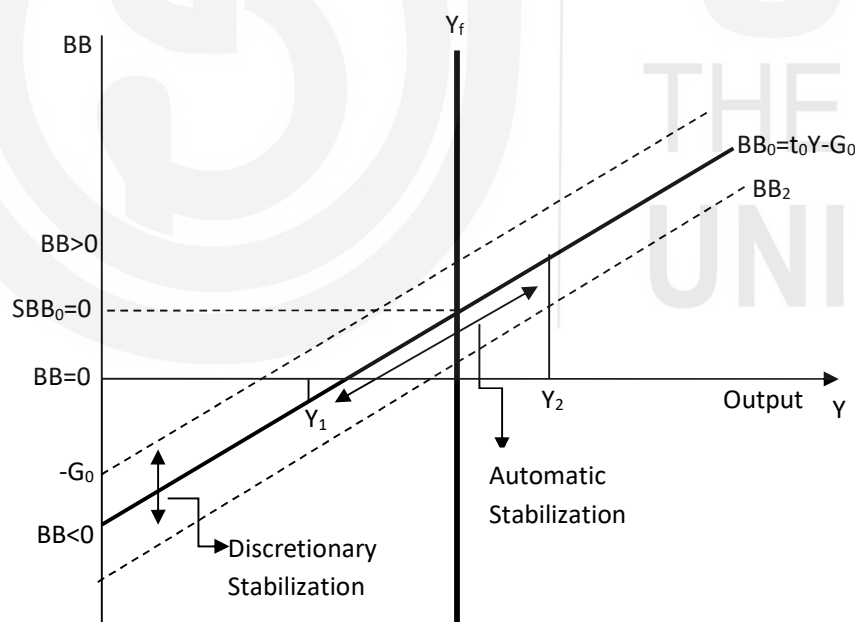


Fig 18.6 shows *Automatic stabilization* comes from changes in the budget balance along the BB_0 line as Y fluctuates between Y_1 and Y_2 . *Discretionary stabilization* shifts the budget function as a result of changes in government expenditure or taxes.

Check Your Progress 3

- 1) What is the difference between discretionary and non-discretionary fiscal policy?

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- 2) What problems does the government face in implementation of discretionary fiscal policies?

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18.5 LET US SUM UP

By fiscal policy we mean automatic and discretionary changes by the government in taxation, public expenditure and borrowing in order to achieve the intended goals of economic growth, full employment, income equality and the stabilization of the economy in its growth path. To carry out these objectives government make use of discretionary and non-discretionary policy measures. Non discretionary policy measures operate automatically and get triggered by the state of the economy. On the other hand, discretionary fiscal policy requires deliberate action by the government to manage aggregate demand. In this chapter we have examined whether policy should take an active or passive role in responding to economic fluctuations. There are many arguments on both sides of these questions. Perhaps the only clear conclusion is that there is no simple and compelling case for any particular view of macroeconomic policy. The existence of various kinds of time lags, or delays in the response of the economy to discretionary policies, can make the economy difficult to control. Automatic stabilizers on the other hand do not suffer from time lags as they are built into the budget program by setting the net tax rate, and work automatically. But automatic fiscal stabilizers only reduce the impact of changes in autonomous expenditure. They do not offset them. Hence, both policy measures should be used in tandem to reduce the fluctuations in output and prices.

18.8 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Fiscal policy is demand management policy. Taxes, transfer payments, and government expenditures on goods and services are the tools of fiscal policy.
- 2) At the time of recession the Government increases its expenditure or cuts down taxes or adopts a combination of both. On the other hand, to control inflation the Government cuts down its expenditure or raises taxes.

Check Your Progress 2

- 1) Refer to Fig. 18.2 and Fig. 18.3

Check Your Progress 3

- 1) By discretionary fiscal policy we mean deliberate change in government expenditure and taxes to influence the level of national output and prices. On the other hand, non discretionary fiscal policy of automatic stabilizers is a built-in tax or expenditure mechanism that automatically increases or decreases the aggregate demand as desired.
- 2) The existence of various kinds of time lags such as recognition lags, implementation lags, and response lags create problems in successful implementation of discretionary fiscal policies.

UNIT 19 FISCAL SUSTAINABILITY*

Structure

- 19.0 Objectives
- 19.1 Introduction
- 19.2 Government's Budget Constraint and Budget Deficit
- 19.3 Current versus Future Taxes
 - 19.3.1 Full Repayment in Year 2
 - 19.3.2 Full Repayment in Year t
 - 19.3.3 Debt Stabilization in Year t
 - 19.3.4 Debt-GDP Ratio
- 19.4 The Dangers of High Debt
 - 19.4.1 Debt Default
 - 19.4.2 Lessons from the Past
- 19.5 Money Finance
- 19.6 Effects of a Decrease in Budget Deficit
- 19.7 Let Us Sum Up
- 19.8 Answers/ Hints to Check Your Progress Exercises

19.0 OBJECTIVES

After going through this unit you will be in a position to

- derive the fiscal sustainability condition from the government budget constraint;
- find out the effects of a fall in taxes and a rise in debt ratio on fiscal sustainability;
- explain the consequences of high debt, debt default and debt monetization; and
- identify the short run and long run effects of a decrease in budget deficit.

19.1 INTRODUCTION

Fiscal sustainability is defined as the government's ability to meet its current and future obligations, to maintain its current policies and satisfy its lifetime budget constraint without defaulting on its debt obligations. It is thus the ability of a government to maintain public finances at a credible and serviceable position over the long term. Ensuring long-term fiscal sustainability requires that governments engage in continual strategic forecasting of future revenues and liabilities, environmental factors and socio-economic trends in order to adapt

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financial planning accordingly. High and increasing debt levels are harmful to governments' fiscal positions and can cause a vicious cycle of growing debt, reducing the potential for economic growth as funds are diverted away from productive investments. One concept of fiscal sustainability relates to a government's ability to indefinitely maintain the same set of policies while remaining solvent. If a particular combination of fiscal and monetary policies would, if indefinitely maintained, lead to insolvency, these policies would be referred to as unsustainable. In case the government cannot service the debt (means repay the principal amount and interest accrued), there is a default on the part of the government. Governments often change their policies if it becomes clear that the government is heading towards a default on its debt. Thus, the focus of fiscal sustainability analysis is not on default in debt servicing, which governments tend to avoid but rather on the policy changes needed to avoid eventual default.

19.2 GOVERNMENT BUDGET CONSTRAINT AND BUDGET DEFICIT

The fundamental building block of fiscal sustainability analysis is the government budget constraint, which is an identity. We must begin with a definition of the budget deficit. One can assess whether fiscal policy is sustainable by examining the size of long-term fiscal imbalances. Doing so provides an early warning of explicit or implicit default and also indicates the magnitude of the adjustments needed to preclude default. The main indicator of a country's fiscal imbalance is its annual official deficit – the difference between annual expenditures (government purchases and transfer payments) and tax and nontax receipts.

We can write the budget deficit in year t as

$$\text{Budget Deficit}_t = rB_{t-1} + G_t - T_t \quad \dots(19.1)$$

All variables are in real terms:

B_{t-1} is government debt at the end of year $t - 1$, or, equivalently, at the beginning of year t ; r is the real interest rate, which we shall assume to be constant here.

Thus,

rB_{t-1} equals the real interest payments on the government debt in year t .

G_t is government spending on goods and services during year t .

T_t is taxes minus transfers during year t .

We can rewrite (19.1) as

$$B_t - B_{t-1} = rB_{t-1} + G_t - T_t \quad \dots(19.2)$$

$$B_t - B_{t-1} = rB_{t-1} - X_t \quad \dots(19.2a)$$

where B_t is the quantity of public debt at the end of period t

$B_t - B_{t-1}$ is the issuance of debt in period t or deficit in period t . If the government runs a deficit, government debt increases. If the government runs a surplus, government debt decreases.

X_t is the primary balance which is equal to $T_t - G_t$

rB_{t-1} is the interest payment on debt where r is the real interest rate

The government budget constraint links the change in government debt to the initial level of debt (which affects interest payments) and to current government spending and taxes. It is often convenient to decompose the deficit into the sum of two terms:

- (i) Interest payment on the debt rB_{t-1}
- (ii) The difference between taxes and Spending ($T_t - G_t$). This term is called the 'primary surplus'.

We can rewrite (19.2a) as

$$B_t = (B_{t-1} + rB_{t-1}) - X_t \quad \dots (19.2b)$$

Intuitively the debt at the end of period t , that is B_t , is equal to the sum of debt in period $(t - 1)$, that is B_{t-1} , and the interest payments on this debt in period $(t - 1)$, that is rB_{t-1} , less the primary surplus in period t , that is X_t . The net debt in period t , $(B_t - B_{t-1})$ is thus equal to the interest payments on the debt in period $(t - 1)$, that is rB_{t-1} , less the primary surplus in period t , that is X_t .

Check Your Progress 1

- 1) Explain the concept of government's budget constraint.

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19.3 CURRENT VERSUS FUTURE TAXES

The sources of funds for a government are limited. There are two sources of revenue for the government, viz., tax revenue and non-tax revenue, of which tax revenue is the major source. Thus, in the analysis that follows we use the term taxes to indicate the total revenue generated by the government. As we all know, there is limited scope on the part of a government to increase tax revenue. The change in taxes depends upon several factors such as growth rate in gross domestic product (GDP), tax base, tax rate, etc.

Let us first consider a one-year decrease in taxes. Let us start from a situation where, until year 1, the government has balanced its budget, so that initial debt is equal to zero. During year 1, the government decreases taxes by 1 for one year. Thus, debt at the end of year 1, B_1 , is equal to 1. Let us consider various possibilities.

19.3.1 Full Repayment in Year 2

Suppose the government decides to fully repay the debt during year 2. From (19.2), the budget constraint for year 2 is given by

$$B_2 = (1 + r)B_1 + G_2 - T_2 \quad \dots(19.3)$$

If the debt is fully repaid during year 2, then the debt at the end of year 2 is equal to zero: $B_2 = 0$. Replacing B_1 by 1 and B_2 by 0 in equation (19.3) and transposing terms gives

$$T_2 - G_2 = (1 + r)1 = (1 + r) \quad \dots(19.4)$$

To repay the debt fully during year 2, the government must run a primary surplus equal to $(1 + r)$. It can be done in one of two ways, viz., (i) a decrease in spending, and (ii) an increase in tax revenue. We shall assume here and in the rest of this section that the adjustment comes through taxes, so that the path of spending is unaffected. It follows that the decrease in taxes by 1 during year 1 must be offset by an increase in taxes by $(1+r)$ during year 2.

19.3.2 Full Repayment in Year t

Now suppose the government decides to wait until year t to repay the debt. So, from year 2 to year $t-1$, the primary deficit is equal to zero—taxes are equal to spending. During year 2, the primary deficit is zero. So, from (19.3), debt at the end of year 2 is:

$$B_2 = (1 + r)B_1 + 0 = (1 + r)1 = (1 + r) \quad \dots(19.5)$$

Where the second equality uses the fact that $B_1 = 1$

With the primary deficit still equal to zero during year 3, debt at the end of year 3 is

$$B_3 = (1 + r)B_2 + 0 = (1 + r)(1 + r)1 = (1 + r)^2 \quad \dots(19.6)$$

Solving for debt at the end of year 4 and so on, it is clear that as long as the government keeps a primary deficit equal to zero, debt grows at a rate equal to the interest rate, and thus debt at the end of year $t-1$ is given by

$$B_{t-1} = (1 + r)^{t-2} \quad \dots (19.7)$$

Despite the fact that taxes are cut only in year 1, debt keeps increasing over time, at a rate equal to the interest rate. The reason is simple: Although the primary deficit is equal to zero, debt is now positive, and so are interest payments on the debt. Each year, the government must issue more debt to pay the interest on existing debt.

In year t , the year in which the government decides to repay the debt, the budget constraint is

$$B_t = (1 + r)B_{t-1} + (G_t - T_t) \quad \dots(19.8)$$

If debt is fully repaid during year t , then B_t (debt at the end of year t) is zero. Replacing B_t by zero and B_{t-1} by its expression from (19.7), in equation (19.8)

$$\text{gives } 0 = (1 + r)(1 + r)^{t-2} + (G_t - T_t) \quad \dots(19.9)$$

Reorganizing and bringing $(G_t - T_t)$ to the left side of the equation implies

$$T_t - G_t = (1 + r)^{t-1} \quad \dots(19.10)$$

To repay the debt, the government must run a primary surplus equal to $(1 + r)^{t-1}$ during year t . If the adjustment is done through taxes, the initial decrease in taxes of 1 during year 1 leads to an increase in taxes of $(1 + r)^{t-1}$ during year t .

This example yields our first set of conclusions:

- 1) If government spending is unchanged, a decrease in taxes must eventually be offset by an increase in taxes in the future.
- 2) The longer the government waits to increase taxes, or the higher the real interest rate is, the higher the eventual increase in taxes must be.

19.3.3 Debt Stabilization in Year t

We have assumed so far that the government fully repays the debt. Let us now look at what happens to taxes if the government only stabilizes the debt. (Stabilizing the debt means changing taxes or spending so that debt remains constant from then on). Suppose the government decides to stabilize the debt from year 2 on. Stabilizing the debt from year 2 on means the debt at the end of year 2 and thereafter remains at the same level as it was at the end of year 1. From equation (19.3), the budget constraint for year 2 is

$$B_2 = (1 + r)B_1 + G_2 - T_2$$

Under our assumption that debt is stabilized in year 2, $B_2 = B_1 = 1$. Setting $B_2 = B_1 = 1$ in the above equation yields

$$1 = (1 + r) + G_2 - T_2 \quad \dots(19.11)$$

Reorganizing, and bringing $(G_2 - T_2)$ to the left side of the equation,

$$T_2 - G_2 = (1 + r) - 1 = r \quad \dots(19.12)$$

To avoid a further increase in debt during year 1, the government must run a primary surplus equal to real interest payments on the existing debt. Taxes are permanently higher from year 1 on, by an amount equal to r : equivalently, from year 1 on, the government runs a primary surplus equal to r . Thus, the following conclusions follow:

- (i) To stabilize the debt, the government must eliminate the deficit.
- (ii) To eliminate the deficit, the government must run a primary surplus equal to the interest payments on the existing debt. This requires higher taxes forever.

19.3.4 Debt-GDP Ratio

We have focused so far on the evolution of the level of debt. But in an economy in which output grows over time, it makes more sense to focus instead on the *ratio of debt to GDP*. Let us derive the debt -GDP ratio first. Rewrite equation (19.2) as:

$$B_t = (1 + r)B_{t-1} + G_t - T_t \quad \dots(19.2b)$$

Now divide both sides of the above equation by real output, Y_t , to get

$$\frac{B_t}{Y_t} = (1 + r) \frac{B_{t-1}}{Y_t} + \frac{G_t - T_t}{Y_t} \quad \dots(19.13)$$

Next rewrite B_{t-1}/Y_t as $(B_{t-1}/Y_{t-1})(Y_{t-1}/Y_t)$. In other words, we multiply the numerator and the denominator by Y_{t-1} .

$$\frac{B_t}{Y_t} = (1 + r) \left(\frac{Y_{t-1}}{Y_t} \right) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.14)$$

Note that all the terms in the equation are now in terms of ratios to output, Y . To simplify this equation, we assume that output growth is constant and we denote the growth rate of output by g , so Y_{t-1}/Y_t can be written as $1/(1 + g)$. And use the approximation $(1 + r)/(1 + g) = 1 + r - g$. Using these two assumptions, let us rewrite the preceding equation (19.14) as

$$\frac{B_t}{Y_t} = (1 + r - g) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.15)$$

Finally, reorganize the above equation to get

$$\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}} = (r - g) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.16)$$

From equation (19.16) we find that the change in the debt ratio over time (the left side of the equation) is equal to the sum of two terms, viz., (i) The first term is the difference between the real interest rate and the growth rate times the initial debt ratio. The second term is the ratio of the primary deficit to GDP. Equation (19.16) implies that the increase in the ratio of debt to GDP will be larger:

- 1) The higher the real interest rate
- 2) The lower the growth rate of output, the higher the initial debt ratio
- 3) The higher the ratio of the primary deficit to GDP

Check Your Progress 2

- 1) What factors cause an increase in the debt to GDP ratio?

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- 2) What must the government do to stabilize the debt?

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